

# UI/UX Design System & Rules Document: Zero-Cost Ephemeral Chat

## Core Design Principles

- **Zero Footprint Visuals:** The UI must visually reinforce ephemerality—no persistent history indicators, no archive drawers, and clear visual cues that closing the app discards the session.
- **Frictionless Entry:** Zero-data onboarding. Only two input fields (Username and Password) on a single screen with immediate transition into the chat.
- **High Contrast & Clarity:** Minimalist interface avoiding heavy assets, custom fonts, or heavy illustrations to maintain lightning-fast load times on free-tier infrastructure.
- **Instant Session Awareness:** Prominent online/offline visual indicators so users know if their peer is currently reachable before transmitting ephemeral packets.

## Color Palette & Typography

### Color Tokens

- **Background Primary:** #0F172A (Deep Slate / Dark Mode Default)
- **Surface / Cards:** #1E293B (Elevated Charcoal)
- **Accent / Action:** #38BDF8 (Sky Cyan — primary buttons, active states)
- **Outgoing Bubble:** #0284C7 (Vibrant Blue — sent messages)
- **Incoming Bubble:** #334155 (Slate Gray — received messages)
- **Text Primary:** #F8FAFC (Off-white)
- **Text Muted:** #94A3B8 (Cool Slate — timestamps, hints)
- **Destructive / Ephemeral Alert:** #F43F5E (Rose / Crimson — exit/wipe session)
- **Status Online:** #22C55E (Vibrant Green dot)

### Typography

- **Primary Font Family:** System Default (-apple-system, BlinkMacSystemFont, Roboto, Segoe UI). Eliminates web-font loading lag.
- **Scale:**
  - Screen Title: 20px, Bold (600)
  - Body / Messages: 15px, Regular (400), Line Height: 1.4
  - Meta / Timestamps: 11px, Medium (500)
  - Input Fields: 16px (prevents auto-zoom on iOS browsers)

## User Flow & Information Architecture

[ Launch App ]

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[ Screen 1: Auth Screen (Toggle: Login / Register) ]

| (Valid Credentials -> JWT Stored in RAM Only)

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[ Screen 2: Direct Chat / Session View ]

├── Top Bar: Target Username Input + Real-Time Status Dot (Green/Gray)

├── Body: Message Feed (Transient React State Array)

└── Bottom Bar: Message Input + Send Button

## Screen Specifications & Layout Rules

### 1. Authentication View

- **Header:** Minimalist logo mark with label Ephemeral Direct.
- **Form Inputs:**
  - Single card centered on screen with two fields: Username and Password.
  - Auto-capitalization: none; Auto-correct: off.
  - Single action button: Enter Room.
  - Text toggle below button: "Need an account? Register" switches mode inline without route changes.
- **Ephemeral Disclaimer:** Subtitle label: "No email. No logs. Closing this window permanently discards all chat history."

### 2. Chat Screen

- **Navigation Header:**
  - Left: User's own handle (@me).
  - Center: Inline text input Chat with @[username] with a presence dot indicator showing recipient socket status.
  - Right: Red icon button Wipe & Exit (clears RAM state, drops socket connection, returns to Auth View).
- **Message Stream Area:**
  - Reverse layout (flex-direction: column with auto-scroll to bottom).
  - Outgoing messages: Aligned right, rounded corners with bottom-right corner squared (border-bottom-right-radius: 2px).
  - Incoming messages: Aligned left, rounded corners with bottom-left corner squared (border-bottom-left-radius: 2px).
  - Transient Timestamp: Displayed inside or below each bubble in 11px muted gray.
- **Input Dock:**
  - Fixed to screen bottom, handles virtual keyboard avoidance natively.
  - Capsule text field with send icon button that enables only when text.trim().length > 0.
  - Hitting Enter sends immediately on desktop browsers; displays "Send" action on mobile keyboard.

## Ephemeral UX State Rules

- **Page Refresh / Exit Warning:**
  - Browser/app event listener triggers standard confirmation modal: *"Chats are not saved. Reloading will erase all messages."*
- **Offline Handling:**
  - If the peer disconnects, an inline system banner appears: @[user] went offline. Sent messages will be dropped.
- **Cold Start Feedback:**
  - Free-tier servers (Render) take 30–50s on initial cold starts. The UI displays an animated loader with copy: *"Waking up server instance (~30s on first load)..."* to prevent user drop-off during dormancy wake-up.

Would you like the complete single-file React Native (Expo) or React.js component code

implementing this design?